

TABLE I
Components and Chips

Manufacturer	Device Family	Part Number of family	Benefits Summary	Applications	Technologies	Key Parameters
Silicon Labs	Wireless Microcontroller	Si106x/ Si108x	Combines high-performance wireless connectivity and ultra-low power microcontroller processing into a small 5mm x 6mm form factor	Home automation Home health care Building HVAC control Telemetry	Compliant with the 802.15.4g smart metering standard	-On-chip debug -Ultra-low power 8051MCU core -2 μ s wake-up time
NXP	Wireless Microcontroller	JN5148	Offers fully compliant 2.4GHz IEEE802.15.4 transceiver, 128kB of ROM and 128kB of RAM to support both the networking protocol stacks	JenNet, JenNet-IP and ZigBee PRO applications	Based on the IEEE802.15.4 standard	-Operating current consumption of 18mA when receiving, and 15mA when transmitting @ +3dBm -Ultra low power consumption -Highly efficient memory architecture -32-bit RISC CPU – 32MIPS
Texas Instruments	Wireless Microcontroller	CC2541-Q1	Enables the building of robust network nodes with low total bill-of-material costs	Both low-energy and proprietary 2.4GHz applications	Bluetooth Low Energy	-2.4GHz low energy compliant -Supports data rates of 250kbps, 500kbps, 1Mbps, and 2Mbps -Excellent link budget, Enabling long-range
Atmel	Wireless AVR Microcontroller	Atmeg- a256RFR2	Combines AVR microcontroller and best-in-class 2.4GHz RF transceiver and offers highest RF performance for single-chip devices.	-ZigBee/IEEE 802.15.4 – full and reduced function device -General purpose 2.4GHz ISM band transceiver with microcontroller -RF4CE, SP100, wirelessHART, ISM applications and IPv6 / 6LoWPAN	ZigBee/IEEE 802.15.4	-Ultra low power consumption (1.8 to 3.6V) for AVR & Rx/Tx: 10.1mA/18.6mA -38 programmable I/O lines -Speed grade: 0 – 16 MHz @ 1.8 – 3.6V range with integrated voltage regulators
Hope RF	Wireless Transceiver Module	RFM12B-S2	Module is implemented with a unique PLL (phase locked loop) and operates in the 915MHz band	Applications compliant with FCC and ETSI regulations		-Wake-up timer -Differential antenna input -Programmable TX frequency deviation (from 15 to 240kHz) -Programmable receiver bandwidth (from 67 to 400kHz)
Silicon Labs	Wireless Transceiver	Si4438 RF	High-performance, low-current, ISM band transceiver covering the 425 to 525MHz frequency band	-Smart metering and automated meter reading -Robust in home communications to home appliances -Long-range backhaul communications to the collector		-Frequency range: 425–525MHz band -Receive sensitivity: –124dBm -Power supply: 1.8V to 3.6V -> 84dB blocking for > 8MHz offset
Texas Instruments	System on Chip	TMS320T-CI6636	Multicore SoC architecture designed for high-performance wireless infrastructure applications		CDMA/HSPA/ HSPA+, TD-SCDMA, GSM, TDD-LTE, FDD-LTE, LTE-A and WiMAX	-Eight TMS320C66x DSP core subsystems per core @ 1.2GHz -ARM Cortex A15 quad core cluster @ 1.2GHz per core with 4MB L2 cache coherent memory
Allen-Bradley	Wireless Interface Module	Bulletin 22-WIM	Provides a wireless communications interface between computers equipped with Bluetooth wireless technology	All products that support the DSI protocol	Bluetooth wireless technology	-Full-duplex point-to-point protocol -Compatible with all powerflex compact-class drives and peripherals